



A Critical Review of Contaminant Intrusion and Dynamics of Herbicide Intrusion in Water Distribution Systems

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Abstract

Public health, infrastructural resilience, and drinking water quality are all persistently threatened by contaminant intrusion into Water Distribution Systems (WDSs). There is mounting evidence that herbicides and their by-products are intruding into Water Distribution Networks (WDNs) across the world. Consequently, owing to the intricate interactions that occur during the intrusion, their prediction and modelling are challenging. The research community has created reaction-transport models and advection-dispersion equations in response to these problems. These typically try to capture part of the reaction kinetics. Furthermore, some of these mathematical models have been integrated into simulation tools, utilising the EPANET software and its EPANET-MSX extension. They have been used in scenario-based simulation studies for demonstrating various interactions between the dynamic states of the parent herbicide compounds and their transformation products. The goal of this review paper is to briefly review the occurrence, mechanism, fate, and significance of pesticide transformation products (PTPs). A review of over 2-decade-and-a-half-long (2000–2026) of journal articles, conference proceedings, technical reports, and experimental, analytical, numerical, and field-based research on contaminant intrusion is presented in this publication. Importantly, this review identifies persistent gaps in the precise fate of herbicides after incursion events into WDS, including pipe material interactions, the impact of residual disinfectants, biofilm-mediated transformations, and hydraulic mixing/dilution effects, with parent chemical transformation into more toxic and persistent metabolites. The paper provides an overview of herbicides and PTPs as contaminants of emerging concern and discusses the modes of degradation of herbicides and pesticides, their properties and associated risks.

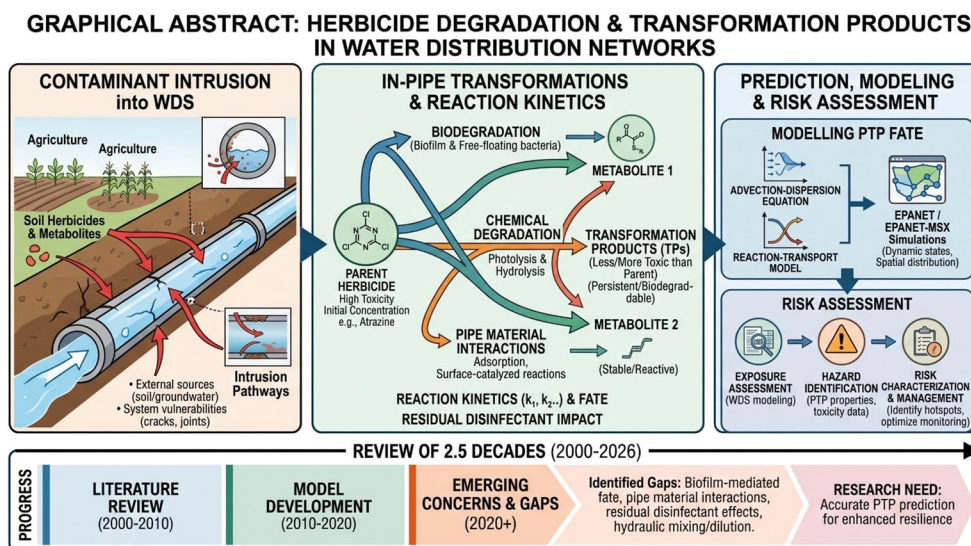
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Graphical Abstract



Keywords Water distribution systems · Contaminant intrusion · EPANET-MSX · Herbicides intrusion · reactive transport modelling

Introduction

A water distribution system (WDS) is the interconnection and assemblage of pipes, hydraulic and flow-control elements, meters and fittings, reservoirs and tanks, and other water reticulation appurtenances for conveying water from treatment plants to end-users at sufficient pressure. WDSs are important in modern and rural communities and are considered critical infrastructure, with public health risks if not well protected against potential contaminant intrusion [1].

The ever-increasing challenges to WDS have long stemmed from growing urbanisation and climate change, which affect water resources [2], and, more recently, the intrusion of contaminants into Water Distribution Networks (WDNs) through potable water pipelines. Contaminant intrusion in a WDS is the incursion of wastewater, pathogens, and harmful industrial and agrochemicals into water supply lines. It is the entrance of contaminants after treatment (for example, through negative-pressure leaks, backflow/cross-connections or permeation). These compromise water quality as it flows to end-users. Contamination between the Water Treatment Plant (WTP) and end users is common and occurs through the distribution pipeline. Water quality deterioration in WDN occurs due to inadequate treatment processes in the WTP [3], deterioration within the WDN [4], and contaminant intrusion [5].

It is crucial to understand that secondary disinfection in WDS is not intended to remove herbicides. Instead of eliminating chemical pollutants, its purpose is to preserve

a residual disinfectant as a microbiological defence. Importantly, the distribution network itself does not contain any treatment units, such as membranes, sophisticated oxidation processes, or granular activated carbon (GAC). As a result, any herbicide that enters the network by intrusion or escapes treatment at the plant will remain there, only to be accidentally transformed by residual chlorine or chloramine. These interactions may result in transformation products that are unregulated and unmonitored but may be more hazardous or persistent than the parent chemical.

Herbicide use in residential garden, urban green space maintenance forms some of the pathways for intrusion (non-point sources), which are aggravated by intense precipitation as a means of transportation to the main water source. These herbicides are mostly unremovable by conventional water treatment processes [6]. Water utilities can better protect public health and handle the intricate, multi-faceted problem of herbicide contamination in water distribution systems by considering the complete process, from application to tap.

This contaminant intrusion can be malicious and deliberate, such as those from a terrorist attack [7]. Others may be accidental due to a deficit or improper WDN operation [4], or pipe leaks that allow contaminants to enter the pipeline during periods of low pressure. This can be lethal. A typical example was the Walkerton outbreak, which claimed the lives of 7 people and made 2,300 people ill due to flooding that introduced *E. coli* into the drinking water supply [4]. Two cases of waterborne illnesses caused by bacterial

infections were reported in Norway and California in 2019 [8]. Several conditions determine the susceptibility of WDN to contamination intrusion. These include pressure variation conditions inside the pipe, the source of contaminant, and the path of intrusion [3, 9]. Water main intrusions can come from contaminated soil, nearby damaged sewers, and contaminated groundwater close to a high-water Tables [9, 10].

In actuality, the examined publications' knowledge gaps exposed serious, observable hazards to public health and water utilities. These gaps include hydraulic mixing, biofilm effects, disinfectant responses, and pipe material interactions. Collectively, these shortcomings indicate that present monitoring – often restricted to grab samples at treatment plant exits and a few network nodes – is severely inadequate. Instead of focusing only on parent chemicals, utilities should immediately switch to real-time monitoring at important nodes found using hydraulic modelling. Conventional treatment approaches must give way to ones that can manage herbicides and their reactive metabolites. Without these adjustments, utilities are successfully controlling herbicide intrusion with insufficient information, putting customers at risk from chemicals they are unaware to test for in each location.

WDS as Vulnerable Infrastructure

WDS are known to be extensive and exposed. This makes them susceptible to physical attack, ageing failure and operational factors like flow velocity, water pressure, cathodic protection, improper operations, and maintenance procedures [11]. Failure of WDS pipelines can happen even when they are properly designed. A backflow event or a direct attack on a reservoir could introduce biological, chemical or radiological agents that can affect a large population long before detection [12]. Any attack or failure of WDS disrupts conveniences, threatens public health, endangers firefighting ability and limits economic activities.

The Growing Threat of Herbicide Contamination in WDS

Herbicides are becoming more frequently found in catchments and plumbing, posing new threats to ecosystems and the safety of drinking water [13, 14]. Atrazine, simazine, metolachlor, glyphosate, bentazone, 2-methyl-4-chlorophenoxyacetic acid (MCPA), and other contemporary hydrophilic herbicides are frequently detected at low amounts (ng/L to µg/L). The ingredient with the highest concentration was glyphosate, followed by 2,4-D and MCPA, and to a lesser extent by dimethomorph, fluopicolide, prothioconazole, and metolachlor (-S), according to [15]. The rationale

for focusing on herbicide intrusion in WDS is due to its persistence, health concerns, and ubiquity [16]. The key gaps in WDS herbicide intrusion control are the lack of regular monitoring of herbicide transformation products at low chronic doses despite their greater persistence and detection frequency, as well as our limited knowledge of combination toxicity and sensitive life stage exposures (newborns, children and adults) [15, 17].

The Evolving Threat Landscape: From Microbial to Chemical Contaminants

Historically, microbial contamination has always been the primary concern in water safety and security, with pathogens causing rapid and visible outbreaks of diseases [18, 19]. Recently, the threat has evolved significantly, with water utilities facing complex and expanding chemical contaminants with chronic and often invisible health risks to our communities. A recent study defines Emerging Contaminants (EC) as newly identified natural or synthetic chemicals or biological agents that are present in the environment and are thought to be potentially dangerous. The emerging chemicals of great concern, such as industrial chemicals, pesticides, herbicides, pharmaceuticals, heavy metals, disinfection by-products, per- and polyfluoroalkyl substances (PFAS), microplastics, etc., are many, unregulated, unmonitored, non-point, and ultimately persistent and bioaccumulative [20]. Long-term exposure typically results in neurological damage, cancer, reproductive harm, endocrine disruption, and chronic health problems. These require advanced analytics and treatment. The evolving threats from microbial to chemical contaminants are driven by industrialisation, pharmaceutical consumption, agricultural intensification, ageing network infrastructures, climate change, etc. The implications for water distribution systems include an increase in intrusion risks through backflow, cross-connection and permeation, persistence in the network, reaction with disinfectants, and monitoring challenges. A common example is Glyphosate (Herbicide), which comes from agricultural runoff and urban weed-control herbicide use, resulting in surface-water contamination before the water gets to the water treatment plants. Another serious risk to the WDS is pipeline infiltration by cross-connection during backflow events. Contaminants can be drawn or forced into the WDS when there is a physical connection between the potable water supply and a possible source of contamination, together with a reversal of normal water flow [21]. Water distribution concerns include chemical persistence in pipes, reactions with chlorine, and difficulty detecting contaminants early and in real time in WDS [22].

Historical Perspectives and Recent High-Profile Contamination Events

Water quality often declines between treatment plants and end-users because of contaminant intrusion along the pipeline, according to earlier water work experts [23]. One of the biggest instances of environmental contamination was the American Flint water disaster of 2014. Flint, Michigan, changed its drinking water supply from Lake Huron water treated and supplied by the Detroit Water and Sewerage Department to the Flint River in April 2014. Numerous issues with water quality and public health resulted from the Flint Water Treatment Plant's improper water treatment. A historic outbreak of Legionnaires' disease occurred in the summer of 2014–2015 due to low chlorine circulation, which encouraged bacterial growth. Also, corrosive water flow degraded the lead pipes of the infrastructure, leading to systemic lead pollution [24]. A devastating waterborne outbreak of the pathogenic germs *Campylobacter jejuni* and *E. coli* O157:H7 occurred in May 2000 in the tiny Ontario village of Walkerton, due to the washing of spread manure from the nearby farm. The spread of the biological contaminant was through the municipal water distribution system, as the chlorinator #5 for one of the wells did not operate properly, leading to a deadly sickness that impacted over 2,300 people and claimed the lives of at least seven citizens [25].

The 4-methylcyclohexylmethanol (MCHM) spill that contaminated the Elk River in West Virginia, USA, on January 9, 2014, is another recent high-profile incident. For four days, 300,000 people's access to drinking water was cut off due to this accident. More than a month after the incident, the distribution network was still being recovered, and the stench of MCHM, which has a very low odour threshold, was still present [26]. There is a compelling argument for ongoing water quality monitoring across the distribution network due to the possible hazards and financial repercussions. For instance, the water in Addis Ababa was found to contain excessive levels of lead, with an average of 17.8 µg/L (above WHO's 10 µg/L guideline), from sources that serve schools. This suggests that not all tap water may be considered "safe" [27].

A study on arsenic contamination in Bangladesh discovered that the mobilisation and release of arsenic into drinking water lines is likely caused by the oxidation of mineral deposits, the release of arsenic from sediments due to reduction, and the release of arsenic from fertilisers and pesticides [28]. According to another study conducted in Algeria, metals in surface water samples most likely came

from a combination of urban and industrial wastes and agricultural practices [29]. A conventional drinking water treatment plant that draws its drinking water from the Tengi River became contaminated due to ongoing and growing use of pesticides in Tanjung Karang, a popular rice-growing area in Kuala Selangor, Malaysia [30]. To guarantee safe drinking water, it is vital to investigate how well the treatment plant removes pesticide contaminants from the water. Advanced treatment methods are required. Water bodies serving treatment plants need regular testing as safety precautions, whereas potable water should undergo advanced water treatment processes as needed.

The Critical role of Modelling in Prevention, Preparedness and Response

Contaminant intrusion into the WDS is frequently localised, fleeting and challenging to identify by traditional monitoring, in contrast to source water pollution. Recent research typically combines modelling and monitoring to obtain a trustworthy representation of the systems and reduce the costs of real-time monitoring [31]. To forecast the probability of water pipe failures and contaminant intrusion fate, a variety of models have been created, which are roughly grouped into statistical, physical models and machine learning [32]. An essential tool for comprehending, forecasting and eventually averting these occurrences is mathematical modelling. Numerous methods for simulating contamination entry are found in the literature. Most methods consider both simulation results and laboratory data.

Optimization studies are typically used to choose monitoring stations, sensor locations, and contaminant source determination [3]. It's critical to identify the best parameters for contamination intrusion modelling. Modelling frameworks were developed by combining underlying network simulation with statistical modelling, classical optimization techniques, evolutionary computation, graph theory, heuristic approaches, greedy approaches, machine learning, and other computational techniques. Three factors, such as conduit characteristics, the driving force, and a contamination source, are necessary for the complicated but commonly occurring contaminant incursion in a WDN. Often, this information is typically incomplete, non-specific and uncertain.

Four major mechanisms for the contaminant's intrusion can be identified. These are leakage pathways, negative pressure events, permeation through the pipe walls and backflow and cross-connection [3, 33]. The role of modelling in understanding these mechanisms is shown in Table 1.

Table 1 Role of modelling in understanding the intrusion mechanism in WDS

Intrusion mechanism	Roles of modelling
Leakage Pathways	Prediction of leak spots and pathways for contaminants when external pressure surpasses internal pressure [33, 34].
Negative pressure events	Hydraulic models simulate transient pressure generated by pump start /stops, valve control, and identify locations with falling pressure below external ground-water levels [35].
Pipe wall permeation	Fate and transport models help in the simulation of organic contaminants like hydrocarbons through plastic pipes based on concentration gradient and properties of the material [1].
Backflow and cross-connection	Network models track flow directions under various demand scenarios, identifying locations prone to flow reversal [36].

Review Scope, Objectives and Search Methods

The scope and goal of modelling the fate of herbicides in water systems (from fields to streams and WDS) are generally constant among research works. The major objectives are:

- Exposure prediction (What concentration reaches different nodes from the source and when [1])?
- Critical hotspot identification (Which area, process and mechanism dominate herbicide dispersion/export [1, 4].

In addition to Google Scholar and ScienceDirect, searches were conducted in scientific databases like Scopus, Web of Science, and PubMed. All the mentioned databases were employed to gather pertinent literature for this investigation and critical review. Some articles were found using the Google search engine. The collected papers were selected according to their applicability to the work’s objectives, considering the year of publication. Relevant articles published from 2000 to 2026, and others that were older but still relevant and served as foundational studies in the proposed modelling works, were used for this review article. Exemplary search strings included combinations such as: (“water distribution system” OR WDS OR “drinking water network”) AND (contamination OR “herbicides”) AND (contaminant intrusion*) AND (Modelling) AND (Herbicides fate under disinfection) AND (herbicide decay).

The primary focused areas of the manuscript are catchment intrusion in water distribution systems, herbicides and pesticides transformation products and lastly the modelling approaches in past studies. All these, with pipe-interaction reactions and interactions, disinfectant effect, biofilm transformation, hydraulic effect, and monitoring inadequacy, are gaps emphasized in the review study.

Physical and Hydraulic Pathways of Contamination Intrusion in WDS

When external contaminants enter WDS through hydraulic or structural weaknesses, usually during low or negative pressure occurrences, this is known as physical intrusion. The majority of research bases intrusion on three concurrent factors: the driving force (pressure differential, typically negative main pressure), the pathway (physical opening), and the contaminant source close to the pipe (such as sewage or contaminated soil) [1, 37]. According to Computational Fluid Dynamics (CFD) and experimental research, intrusion starts when the internal head falls below the exterior head, and the rate of intrusion is influenced by surrounding media, opening shape, and pressure [38]. Reviews highlight how pressure dynamics and physical channels combine to generate a complex intrusion process; even little leaks can pose a serious health concern when they coincide with transitory occurrences and adjacent sewage sources.

Contaminants outside the pipe infiltrate through the pipe wall or gaskets into the drinking water by permeation, a material-driven incursion pathway, even when the pipe is not obviously leaking. It primarily affects plastic and polymeric pipes that are exposed to wildfires or placed in contaminated soils. Diffusion-partition models imply that PE/PVC pipes are at the highest risk in contaminated soils. The partition (solubility) coefficient, external concentration/activity, pipe wall thickness and diameter, and contaminant diffusivity in polymer are important factors [39]. One of the distribution system’s biggest threats to public health and a frequent cause of epidemics is cross-connections and backflows [40]. Only when hydraulic conditions momentarily “pull” or “push” impurities into the pipe may they enter through leaks, cracks, valves, or cross-connections. These occurrences are primarily associated with transitory (unsteady) flow behaviour and low or negative hydraulic conditions, particularly during transients brought on by pump trips, valve operations, or pipe breaks acting on pre-existing leaks or openings, which allow WDS infiltration. Higher baseline pressures, rapid pressure stabilisation, and effective surge/leak control significantly lower risk, whereas the depth and length of negative pressure, the network zones impacted, and leak/valve characteristics determine the severity of incursion [41].

Synthesis of Existing Studies and case Applications

Most of the research on herbicides and drinking water concentrates on source waters and how utilities and catchment management react when herbicides get into supply lines. Some studies say that treated drinking water itself may

contain herbicides such as glyphosate in southern Brazil drinking water [42]. Acid and triazine herbicides (like atrazine, MCPA, simazine, terbutylazine and metolachlor) are often detected in streams, reservoirs, and private wells used as sources of drinking water in agricultural catchments in South Africa, frequently above or at the EU limit of 0.1 µg/L for individual pesticides [43].

The EU value of 0.1 µg/L for individual pesticides and 0.5 µg/L for total pesticides is a precautionary, policy-based compliance standard which is a deliberately stringent value designed to minimize exposure to synthetic pesticides irrespective of their individual toxicity [44] (London et al., 2000). This is different from health-based guideline values derived from toxicological data and incorporate safety margins to take care of uncertainty and vulnerable populations. While the US EPA and WHO use compound-specific health-based limits that vary widely by agent toxicity, the EU/UK approach sets a consistent, low “compliance” value for all pesticides. The WHO/EPA regulations seek to specify a level at which no significant health risk is anticipated, but the EU/UK standard is purposefully conservative. It has been said that South Africa’s drinking water pesticide regulations are inadequate, with few monitoring data, few standards, capacity constraints, and no transformation product coverage [45]. Studies indicate that pesticide detections in South African water sources are frequently above the EU 0.1 µg/L parametric value but remain below WHO/EPA health-based limits, underscoring the disparate outcomes of implementing different regulatory philosophies [44].

Analytical target lists for common herbicides vary from the EU to US EPA and South Africa. EU covered over 100 pesticide parent compounds, with limited coverage of pesticide transformation products (PTPs) [46]; the US EPA has an Unregulated Contaminant Monitoring Rule (UCMR) but includes some pesticides and PTPs with other evolving lists [47], while South Africa has no comprehensive list and limited routine monitoring and minimal coverage.

Due to their chemical and physical properties, polar, non-volatile herbicides like glyphosate and AMPA are particularly difficult to analyse [48]. Some of the key analytical parameters for herbicides include the Limit of detection (LOD)- lowest concentration distinguishable from blank (e.g., < 0.1 µg/L for EU compliance; Limit of Quantification (LOQ)- lowest concentration that can be quantified with acceptable precision (typically 2–5 × LOD); recovery (percentage of analyte recovered from spiked samples; matrix effects (interference from water matrix like ions and organic matter; and sample preparation (methods to concentrate/purify analytes). These parameters suggest analytical techniques like LC-MS/MS (triple quadrupole), applicable in routine quantification of polar herbicides and PTPs [49]

or HRMS (Q-TOF, Orbitrap) for Non-target screening and identification of PTPs and degradation products [48] and other analytical methods.

One modelling study uses EPANET/EPANET-MSX to explicitly simulate pesticide intrusion into water distribution systems and its interaction with chlorine. It then uses machine learning (kNN, random forest) to locate the source of contamination, achieving high localisation accuracy in scenarios involving synthetic pesticide intrusion [26]. The source-pathway-driving force triad with nearby chemical sources (such as contaminated groundwater or agricultural drains), negative pressure, leaks or breaks in the main is highlighted in broader intrusion reviews [50]. This is because true “intrusion” into distribution mains is primarily addressed via modelling and detection algorithms rather than documented field cases.

Key Findings from the Herbicides’ Intrusion in WDS Studies and Knowledge gaps

There are comparatively few studies on actual pipe network “intrusion,” with most concentrating on herbicides travelling from catchments into sources and finished drinking water. When combined, they exhibit a few recurring patterns.

- From source to tap, herbicides are commonly used.
- Pesticide transformation products (PTPs) are often more problematic than the parent compound or its active ingredients.
- Intrusion is often controlled by factors like sources, concentration at the sources, proximity of the source to the point of entry, pipeline materials and hydraulic flow within the WDS.
- There is a serious need to create a model that will predict the fate of herbicides in WDS.

Overall, research clearly documents herbicides in catchments and drinking water sources. Still, there is a major gap in empirical and modelling work that follows herbicides through treatment and real distribution system intrusions to the consumer’s tap. Most studies stop at the catchment and source water scale; true intrusion into distribution pipes and tap-level exposure is poorly investigated. Particularly for episodic peaks and “shock loads” of mobile, polar herbicides, there is insufficient data to connect catchment concentrations to what enters treatment plants and then the network. Although commonly known, transformation products (such as those of flufenacet, metazachlor, and glyphosate) are nevertheless inadequately tracked along the entire source-treatment-distribution cycle and are not monitored in many drinking-water systems. There are very few field case studies that validate such models in actual incursions;

one of the studies specifically simulates pesticide/herbicide intrusion into water distribution systems and examines data-mining techniques to locate the source [26]. Important gaps exist in soil-pipe interactions, realistic intrusion rates, leak geometry, and the fate of herbicides after they enter ageing chlorinated pipe networks [26, 51]. It is still difficult to quantify mixture toxicity, long-term low-dose exposure, and interactions with other stressors in drinking water [52].

Governing Equations for Contaminant Transportation and Transformation in WDS

Several equations are adopted for the description of transportation and transformation of contaminants in WDS, and these include [53]:

- a) One-dimensional Advection-Dispersion-Reaction (ADR) Equation, which is the fundamental mass balance for a contaminant in a pipe under unsteady, one-dimensional flow represented by:

$$\frac{\partial C}{\partial t} = -u \frac{\partial C}{\partial x} + D \frac{\partial^2 C}{\partial x^2} + R(C) \quad (1)$$

where C is the contaminant concentration (mg/L), t is time (s or day), u - average flow velocity (m/s), x - distance along the pipe(m). This was simplified by the standard EPANET to advection-reaction (neglecting dispersion) as;

$$\frac{\partial C}{\partial t} = -u \frac{\partial C}{\partial x} + R(C) \quad (2)$$

- b) Junction and Tank Mixing Assumption: The components, the mixing assumptions and the mathematical expression are stated as follows in Table 2:

Default in EPANET Complete mixing for tanks and junctions. EPANET-MSX allows user-defined tank mixing models.

1. Wall Mass-Transfer and sorption Terms: Herbicides can adsorb to corrosion scales, biofilm, and pipe walls.

Linear equilibrium is assumed in a basic reversible sorption model:

where S is the Sorbed concentration on the pipe wall (per unit volume of pipe), K_d - distribution coefficient (sorption constant).

The next flux to the wall is:

$$J_w = K_w * (C - C_{eq}) \quad (4)$$

where J_w is the mass flux to the wall, K_w - wall mass transfer coefficient, and C_{eq} is the equilibrium concentration of the wall.

In EPANET- MSX, wall species are defined with the WALL keyword, and reactions are expressed using;

$$RATE \text{ species} - K_{wall} * (C \text{ bulk} - C_{eq}) \quad (5)$$

- d) Reaction Expressions

- i) First- Order Decay: Used for many herbicides, including glyphosate degradation in bulk water:

$$R(C) = -K_1 (C) \quad (6)$$

where K_1 is the first-order rate constant (day^{-1}):

$$\text{Half - life} : t_{1/2} = \ln(2) / K_1 \quad (7)$$

- ii) Second-Order Reaction

For reactions where decay depends on two species (e.g., glyphosate+chlorine) as in WDS with disinfectants:

$$R(C) = -K_2 * C \text{ herbicides} * C \text{ chlorine} \quad (8)$$

where K_2 is the Second -order rate constant (L/mg. Day).

Example (glyphosate-chlorine):

$$R(\text{Gly}) = -K_{Cl} [\text{Gly}] [\text{Cl}_2] \quad (9)$$

- iii) Reversible Sorption Model

Table 2 Junction and Tank Mixing Assumptions with mathematical expressions

Components	Mixing assumption	Mathematical expression	Notes
Junction	Instantaneous complete mixing	$C_{out} = \sum Q_1 C_1 / \sum Q_1$	All inflow mixes instantly at the nodes; no storage volume
Tanks	Fully mixed (ideal)	$V \frac{\partial C}{\partial t} = Q_{in} C_{in} - Q_{out} C - kCV$	Assumes uniform concentration throughout.
Tanks	Plug-flow (piston)	$\frac{\partial C}{\partial t} = -u \frac{\partial C}{\partial x} + R(C)$	Can be approximated by multiple complete-mix segments in series.
Tanks	Two-compartment mixing	$V_1 \frac{\partial C_1}{\partial t} = Q(C_{in} - C_1) + D(C_2 - C_1)$	Models short-circuiting and dead zones; used in EPANET- MSX

First-Order reversible sorption

$$\frac{\partial S}{\partial t} = K_{ads} C - K_{des} * S \quad (10)$$

where K_{ads} is the Adsorption rate constant (day^{-1}), K_{des} is the Adsorption rate constant (day^{-1}).

e) Time-Varying Intrusion Boundary Conditions

An intrusion event is represented as a time-varying mass source at a node. The general form is:

$$C_{intrusion}(t) = C_{source} * f(t) * \delta(x - x_0) \quad (11)$$

where C_{source} is the concentration of contaminant in the external source (e.g., contaminated groundwater), $f(t)$ is the time-varying function (e.g., step, pulse or transient-driven), and $\delta(x-x_0)$ is the Dirac delta indicating injection at a specific node.

Example Functions

Step : $f(t) = 1$ for $t_0 \leq t \leq t_1$

Pulse : $f(t) = \exp[-k(t - t_0)]$ (exponential decay)

Transient – driven :

$f(t) = Q_{intrusion}(t) \cdot C_{source}$, where $Q_{intrusion}$ is derived from transient pressure simulations.

Framework for Making Decisions when Choosing a Solver

The hydraulic regime, data accessibility, and chemistry complexity all influence the solver selection. It is always good to start with available data, then move to unknown data. Table 3 presents how the three parameters above can be combined in the selection of an appropriate EPANET extension for solving contaminant problems in WDN.

Transport Dynamics of Contaminants in WDN

The velocity field from hydraulic analysis drives most models, which use 1-D advection–dispersion–reaction equations along pipes to depict contaminant flow [58]. Advection is bulk flow transfer that is managed by network hydraulics, demand trends, and operational modifications (valve/pump operations) [58, 59]. Diffusion and dispersion: Molecular diffusion and velocity gradients disperse pollutant plumes, particularly in laminar or transitional regimes and at low velocities; ignoring these effects can distort the structure

Table 3 Decision framework for Solver Selection for WDN contamination

Hydraulic regime	Data available	Reaction complexity	Recommended EPANET extension with citations
Steady or slowly varying	Good demand / pressure data.	Single species, first-order decay.	Standard EPANET. [53, 54].
Steady or slowly varying	Good demand / pressure data.	Multi-species, complex reactions.	EPANET-MSX. [53].
Steady or slowly varying	Uncertain pressure; low-pressure zones.	Single or multi-species.	Pressure-Driven demand (EPANET-PDX + MSX). [55].
Rapid transients (pump trips, valve closure)	Transient pressure/flow data available.	Single species	EPANET coupled with Transient model (HAMMER, TSNnet). [56].
Rapid transients	Limited transient data; only steady-state available.	Multi-species	Use steady-state estimate of intrusion volume; then standard EPANET-MSX for transport. [57].
Fully dynamic (transient intrusion +multi-species)	High-resolution SCADA; validated transient and water quality models.	Multi-species+ wall sorption	Fully coupled transient+EPANET-MSX (custom coupling). [1].

and persistence of plumes [60, 61]. Reactions and wall interactions: single- or multi-species reactive transport formulations are needed because pollutants may decay, react (e.g., disinfectant–DBP formation), or sorb/desorb at pipe walls [60].

Frameworks for Modelling

Classical tools such as EPANET use a Lagrangian transport algorithm assuming primarily advective plug flow, often treating dispersion in a simplified way [1, 58]. Current advancements expand on this by:

- using random-walk solvers (EPANET DD) (dynamic dispersion) to implement advective-diffusive-dispersive transport with dynamic flow and 2D effects to better capture double peaks and improve spreading under variable regimes [61].

- including reactive multispecies models (EPANET MSX, EPANET C, and EPyT C) that simulate/replicate the linked fate of disinfectants, by-products, and other chemicals under hydraulic conditions and realistic, time-varying demand [1, 62].
- In addition to typical 1D assumptions, computational fluid dynamics (CFD) is used to investigate scenarios where local mixing and 3D effects significantly influence non-reactive pollutant dispersal [63].

Role of Demand Variability and System Dynamics

Patterns of consumer demand influence the velocity field, which in turn influences the placement and dilution of plumes. Changes in consumer behaviour during an event can dramatically influence flow directions, plume trajectories, and exposure, according to stochastic demand models and agent-based frameworks [59]. Long residence times and increased dispersion cause significant spatial variability in concentrations in dead ends and low-flow zones [60].

Emerging Directions

To simulate the fate of contaminants, including emerging contaminants, in real time, new digital twin concepts combine stochastic demand models, sophisticated hydraulic solvers, and flexible water-quality engines with sensor data assimilation to enhance state estimation and source tracking [1, 62]. In general, transport dynamics in WDNs show a tightly coupled system in which human behaviour, hydraulics, dispersion, and reactions all work together to decide when, where, and at what concentrations contaminants reach consumers. The core mechanisms and modelling tools for contaminant transport in WDNs are shown in Table 4.

Table 4 The core mechanisms and modelling tools for contaminant transport in WDNs

Aspect/mechanism	Brief description	Citations
Diffusion/Dispersion	Spreading due to velocity gradients, which is very common at low flow.	[61, 63].
Reaction/ wall exchange	DBP formation, sorption or desorption, Decay.	[1, 58].
Advection	Flow-driven movement along pipelines.	[1, 58, 59].
Lagrangian 1D models	Standard EPANET transport approach.	[1, 58].
Multi-species tool	EPANET-C, MSX, EPyT-C (<i>EPANET-Python</i>) for coupled fate.	[1, 62].
Digital twins	Transport modelling and Real-time, sensor -driven fate.	[62, 64].

Advection and Dispersion in a pipe Network

In WDS, advection and dispersion are important mechanisms controlling the movement of solutes, including herbicides. The bulk movement of water carrying dissolved materials is referred to as advection, whereas the spreading of these materials because of changes in velocity and molecular diffusion within pipes is referred to as dispersion. Because empirical methods frequently underestimate dispersion coefficients under various flow regimes, which affects predictions of contaminant concentration profiles, accurate modelling of longitudinal dispersion in WDS is difficult [1]. Herbicide spread and persistence in water networks are thus influenced by the interaction between advection and dispersion, which may result in localised contamination or dilution depending on flow conditions. Understanding these pathways is important for controlling the hazards of herbicide incursion in WDS and enhancing water quality models to more accurately forecast contaminant behaviour [1].

Herbicide Transformation During Treatment Stage

Herbicides may change during drinking water treatment before water reaches the distribution network. Polar herbicides can only be partially removed by processes such as coagulation, flocculation, sedimentation, and filtration [65, 66]. Through reactive oxygen species (ROS) like hydroxyl radicals ($\text{OH}^\cdot$) and sulfate radicals (SO_4^{2-}), advanced oxidation processes (AOPs) such as UV/ H_2O_2 , UV/ TiO_2 photocatalysis, ozonation, electro-Fenton, and activated persulfate/peroxymonosulfate can degrade herbicides. However, unless built explicitly as in-network booster systems, these procedures are typically missing from distribution pipes and are purposefully designed as treatment units. The consequences for WDS are that the designed systems' rate constants, (e.g., $k_{\text{OH}} = 10^8\text{--}10^{10} \text{ M}^{-1} \text{ s}^{-1}$ $\text{kOH} \approx 10\text{--}10^3 \text{ M}^{-1} \text{ s}^{-1}$) are not transferable to distribution networks due to significant differences in pH, oxidant dosage, water matrix, and contact time. These processes don't happen in pipes unless a utility deliberately doses an oxidant into the network.

Reactions in WDS with Residual Disinfectants (Chlorine, Chloramine)

The specific behaviour of herbicides during intrusion events in WDS, where conditions may differ from open aquatic ecosystems due to pipe materials, residual disinfectants, biofilms, and hydraulic dynamics, is still poorly understood [67]. The main oxidant that is available once water reaches the distribution system is residual chlorine

or chloramine. These residuals may react with herbicides, although the responses are usually less thorough and slower than those of designed AOPs [68, 69]. In contrast to chloramine, which reacts more slowly and may produce nitrogenous disinfection by-products (DBPs) like NDMA, glyphosate combines with free chlorine to produce AMPA and other breakdown products. Under chlorination, atrazine and other triazines undergo dealkylation and dechlorination, creating metabolites that could be more hazardous or persistent [70]. The second-order rate constants for chlorine–herbicide reactions (e.g., $k_{app} = 2\text{--}288 \text{ M}^{-1} \text{ s}^{-1}$) should be used with caution. Only certain pH, temperature, and matrix conditions allow their use. It is not suitable to transfer directly to a WDS model without first confirming the same pH, chlorine dose, and contact time. Microbial degradation, photolysis, hydrolysis, advanced oxidation processes (AOPs), and adsorption to organic matter or pipeline materials are the primary processes that affect herbicide decay. Different from open waters, pipe surfaces and biofilms can either stimulate or hinder transformation reactions [65, 71, 72]. For example, the biodegradation of atrazine varies greatly depending on temperature, pH, microbial activity, light exposure (photolysis), and organic matter concentration [73]. Bacterial breakdown is the main way that many herbicides dissipate; fungi play a smaller role [74]. For many chemicals, degradation rates follow first-order or pseudo-first-order kinetics [68, 71], with half-lives varying from hours under ideal AOPs to months or years under dark/anoxic circumstances [75, 76]. Disinfectants may change or partially break down pollutants, but they may also produce hazardous by-products [65, 71, 72]. Pesticide Transformation Products (PTPs) can be more mobile or hazardous than parent molecules, so degradation does not always produce harmless byproducts according to Anagnostopoulou et al. [77] and [78]. In WDS, oxidation and chlorination processes mainly lower the concentration of parent herbicide chemicals, but this does not translate into a decrease in danger. Oxidation of glyphosate produces AMPA, a metabolite that is more persistent and equally hazardous. Dealkylated derivatives of atrazine might still have endocrine action. Under typical WDS disinfectant residuals (0.2–0.5 mg/L Cl_2 , variable pH, and presence of NOM), complete mineralization—the conversion of organic carbon to CO_2 is rarely accomplished. As a result, it is crucial to track both parent and product species using a mass-balance technique. Instead of evaluating the parent alone, toxicity endpoints should be evaluated for the total of the parent and transformation

products. Without proving similar pH, dosage, matrix, and contact circumstances, rate constants from laboratory research (such as UV/ H_2O_2 , electro-Fenton, and high chlorine doses) shouldn't be immediately used in WDS models.

Dark Reactions and Hydrolysis in WDS

Herbicides may experience dark reactions and hydrolysis (reaction with water) when there is no light and little oxidant residue. At $\text{pH} > 8$, several herbicides, including atrazine and other triazines, undergo base-catalysed hydrolysis, which can happen in pipelines lined with cement. Although this is less frequent in drinking water, sulfonylureas hydrolyse more quickly at low pH [70]. In distribution networks with lengthy residence durations, these abiotic processes contribute to long-term modification despite their modest pace (half-lives of weeks to months). Under dark conditions, biotic processes like microbial breakdown in biofilms can also take place [39]. Implication for WDS is that, although slow, hydrolysis and dark reactions are important for herbicides with extended hydraulic residence durations (such as dead ends and storage tanks). Only when their residence period is longer than the average decay half-lives can they be included in fate models.

Herbicides Adsorption into pipe Walls and Biofilms

Scales (mineral deposits) and biofilms (microbial layers) inside pipes can both adsorb and subsequently release pollutants, changing the water quality over time. Pipe biofilms, which are complex microbial layers combined with extracellular polymeric substances (EPS), can interact significantly with herbicides that enter drinking water or distribution systems [79]. Particularly in old or corroded networks, biofilms and scales may exfoliate, releasing bacteria and sorbed pollutants. Biofilm formation and sorption behaviour are significantly influenced by pipe material, corrosion, stagnation, and disinfectant levels. Biofilm mass and its function as a contamination reservoir are both affected by control techniques (disinfection, flushing, corrosion management) [80]. As a sink and delayed source, biofilms can hold onto herbicides even after they are no longer visible in the bulk water. Sorption frequently exhibits significant potential accumulation capacities in the biofilm matrix and is non-linear (e.g., Langmuir-type for diuron) [80].

Modelling Approaches for Contaminant Intrusion and Transport Within WDS

Modelling approaches for contaminant intrusion in WDS have always sought to estimate where and how much pollution enters, as well as anticipate its fate and travel across the network. This is done within a water quality modelling framework, which is a structured set of tools and methods used in the simulation of the way pollutants move, transform and are removed in water distribution systems in space and time. To reduce numerical dispersion and computation time, transient hydraulic models (Eulerian) combined with Lagrangian water-quality solvers are frequently employed to simulate intrusion during pressure transients and subsequent transit [23]. Compared to constant flow assumptions, some models improve realism by directly linking intrusion rate to local pressure and surrounding soil properties [37]. Others consider terrain slope effects on spread and simulate intrusion from sewers or wastewater [5].

Risk frameworks describe intrusion potential or risk across networks by combining pathway, source, and driving force information using fuzzy logic, Dempster-Shafer theory, or fuzzy-rule systems. Critical nodes are ranked, and propagation risk is evaluated using EPANET simulations and complex network theory. Modelling for “preparedness–detection–management” is organised by broader reviews, which emphasize uncertainty in contaminant kind and source [81].

EPANET and Extended Period Simulation

Hydraulic-water quality models, which monitor how a pollutant enters, travels, mixes, and decays over time, are typically utilised to investigate the intrusion of contaminants into WDS. The main open-source program for this is called EPANET, and it uses extended period simulation (EPS) to reflect variations in demand, operation, and water quality. With inputs like source location, injection pattern, and reaction coefficients, EPANET 2.2 simulates the movement of contaminants (such as chemicals, pesticides, and chlorine) under hydraulic conditions [61].

Standard EPANET tracks one chemical species (or water age/source tracing) at a time, while EPANET-MSX represents multiple interacting bulk and wall species. EPANET-MSX uses BULK and WALL phases; wall species are surface-associated rather than simply ‘attached and mobile.’ Other EPANET tools described in previous works include EPANET-C, EPyT-C, EPANET-DD, and EPANET-PMX. They are not interchangeable elements of

the main program; rather, they are unique, research-focused extensions or wrappers built around the standard EPANET hydraulic engine. An “umbrella simulation platform” called EPANET-C was created to make using EPANET-MSX to represent contamination occurrences in WDS easier [82]. Its main goal is to make multi-species water quality modelling more accessible to people without programming experience by overcoming the complicated user interface and programming requirements usually associated with EPANET-MSX.

In order to do this, EPANET-C offers pre-built and editable directories of conceptual and mathematical models that explain chemical, biological, and physical reactions. It combines EPANET and EPANET-MSX, simulating various contamination situations with a straightforward command-line interface. In 2022, it was presented at the World Environmental and Water Resources Congress.

A completely independent multi-species reactive transport modelling plugin for EPyT (the EPANET-Python Toolkit) is called EPyT-C (where “C” refers to contaminant). It was created because the regular version of EPyT only allows for single-species water quality analysis.

To perform multi-species reactive transport modelling, EPyT-C uses an integrated water quality solver and invokes the hydraulic solver from EPANET 2.2 [62]. It offers pre-configured modules that specify the governing equations for intricate water quality factors, like the “Bacterial regrowth” and “Chlorine decay and Trihalomethanes formation” modules. Numerical accuracy, response coefficients, and input modification are all possible with this tool. “EPANET-DD” is the name of EPANET’s Research Extension for Pressure-Driven Demand. It might allude to an application of pressure-driven demand (PDD) analysis tailored to a particular study. Other extensions, like WNTR (the US EPA’s Water Network Tool for Resilience), provide the option to perform hydraulics using PDD simulations. Additionally, it is used in the EPANET-PDX extension and EPANET-PMX [83]. An integrated network analysis model called EPANET-PMX (Pressure-dependent Multi-species eXtension) was developed to overcome the shortcomings of current tools, mainly their incapacity to perform pressure-driven analysis in conjunction with multi-species water quality simulations. It expands on the work of EPANET-PDX, a pressure-driven extension of EPANET 2 that maintained water quality modelling for a single species but was unable to simulate several chemical compounds simultaneously [55]. This is addressed by EPANET-PMX, which combines the multi-species modelling capabilities of EPANET-MSX with pressure-dependent demand analysis. This enables more accurate low-pressure simulations, which are essential for evaluating

system performance and contamination hazards during incidents like pipe bursts, power outages, and firefighting.

To predict affected nodes, consumer doses, and the time of plume arrival, it is frequently used to model purposeful or accidental contamination [61]. Advanced applications combine EPANET with AI or optimization for source identification and risk management, such as data mining to locate pesticide incursions modelled with EPANET/EPANET-MSX or Bayesian optimization to locate intrusion sources in near real time [84]. Run in extended-period simulation mode, EPANET is a key instrument for evaluating contaminant incursion in WDS. It supports source identification, long- and short-term plume tracking, risk assessment, and consequence management, all of which are frequently improved by multi-species and sophisticated hydraulic extensions. Multi-species reactions and more realistic contaminant chemistry during intrusion events are made possible by EPANET-MSX, EPANET-C, and EPANET-PMX [1]. Reliable intrusion modelling depends on the stability and accuracy of extended period simulation (EPS), which is improved by new numerical techniques (e.g., better Lagrangian/event-driven transport, EPS/GGA, DD\EPANET) [85].

The path from a contamination source to a customer's tap in a water distribution system involves several steps. Accurate modelling calls for unique skills that deal with various physical processes. While standard EPANET offers a good starting point for many tasks, it is inadequate for simulating the entire intrusion pathway, particularly the dynamics of contaminant penetration during an event such as water hammer, because it relies on significant simplifications. Table 5 lists the five key capabilities required, their differences from a standard EPANET, and the tools that can be used to put them into practice.

In reality, these capacities must be coupled for a true mechanistic simulation of contaminant infiltration. For example, a model may mimic a transient pressure wave using a technology such as TSNNet (Capability 3). The rate of contaminant penetration would then be ascertained by feeding the pressure data into an intrusion-flow calculation (Capability 4). Lastly, EPANET-MSX's ADR solver (Capability 5) would be used to mimic this pollutant source to monitor its transport and transformation along the network. This methodical, comprehensive approach is necessary for a precise assessment of the risk.

EPANET-MSX for Multi-Species Reactive Transport

EPANET-MSX, as an extension of EPANET, enables explicit simulation of multiple interacting chemical species in drinking WDN, capturing interactions at pipe walls and in bulk water [89]. With simple mathematical expressions,

Table 5 EPANET Key capacity requirements with their differences from standard EPANET and implementation tools

Capability	Function	How Standard EPANET Differs	Software / Method (with Citation)
1. Quasi-Steady EPS	Simulates hydraulic and water quality behaviour over time (e.g., 24 h) by assuming conditions are stable at each calculation time step.	This is the core functionality of Standard EPANET. It is the baseline method for network analysis.	EPANET 2.2 (US EPA).
2. Pressure-Driven Demand (PDD) Analysis	Uses a function to reduce actual water delivery at a node when the available pressure is insufficient, modelling "pressure-deficient" conditions.	Assumes demand is always fully met, regardless of pressure. This can overestimate system resilience and understate intrusion risks during low-pressure events.	EPANETpdd (University of Exeter). [86, 87].
3. Transient / Water Hammer Models	Simulates rapid pressure changes (surges) caused by events like pump trips or fast valve closures, which are the root cause of many intrusion events.	Cannot simulate these rapid, unsteady hydraulic events, as it is designed for slower changes.	TSNet (Open-Source), WNTR (US EPA), MIKE+(DHI). [88].
4. Intrusion-Flow Equations	Calculates the actual volume of contaminated water that enters through a leak, driven by the pressure difference between the outside environment and the inside of the pipe.	A contaminant injection is typically defined by the modeller as a fixed input (a source), not as a dynamic result of the system hydraulics.	Custom Code / WNTR. Requires coupling with a transient model and using equations like the orifice equation. [37].
5. Subsequent Water Quality Transport	Simulates how the intruded contaminant moves through and reacts within the pipe network, including processes like advection, dispersion, and reaction.	Standard EPANET can model advection-reaction (movement and decay) but ignores longitudinal dispersion.	EPANET-MSX (US EPA) can solve the full Advection-Dispersion-Reaction (ADR) equation.

EPANET-MSX enables users to construct any number of species and specify arbitrary reaction networks (kinetic and equilibrium) [90]. It combines stiff ODE solvers for reactions across the network with EPANET's hydraulics and a Lagrangian advection–reaction algorithm to solve transport [91]. Realistic modelling of adsorption, biofilm/wall interactions, and multi-source water mixing is made possible by

Table 6 Typical applications of EPANET-MSX

Area of Application	Processes modelled	Citations
Exotic reactions/ Security	Dispersion and Nicotine-chlorine reaction with the reactive intermediate	[90].
Disinfection and Disinfection By-Products	Chlorine decay, chloramine, HAAs/ THMs, multi-source mixing	[61, 89].
Odour and Taste	Multi-species model for 2,4,6-TCP to 2,4,6-TCA formation	[1].
Contaminants and Metals	Arsenite adsorption in chlorinated systems, Arsenite oxidation	[90].

the ability of species to be either attached (wall-phase) or mobile (in the flow) [89]. Table 6 shows typical applications of EPANET-MSX.

Parameterization of Herbicide Reaction Kinetics

Parameterising herbicide kinetics means transforming observed degradation, sorption, and transport behaviour into numerical rate constants and model forms that can be used in fate and risk models. Rate laws (first- or second-order) and their reliance on pH, oxidant dose, light, and matrix components are commonly used to characterize the kinetics of herbicide reactions in water distribution (or treatment) systems. Experts can forecast herbicide degradation and byproduct development along treatment trains or pipes using these factors. With apparent rate constants that change with pH and species (HClO vs. ClO⁻), direct reactions with disinfectants such as chlorine can exhibit second-order kinetics [70]. Sulfonylureas and chlorine are frequent examples of these reactions: Second-order $k_{app} = 2\text{--}288 \text{ M}^{-1} \text{ s}^{-1}$; half-lives $\sim 18\text{--}27 \text{ min}$ at pH 7 and 4 mg/L Cl₂ [70]; k_{O_3} varies from ~ 2 to $2190 \text{ M}^{-1} \text{ s}$ for phenylureas with ozone/OHUV/PS promotion: pseudo-first-order $k_{obs} \approx 0.236 \text{ min}^{-1}$; electro-Fenton 2,4-D derivative: $k \approx 0.47 \text{ min}^{-1}$ [92]; k_{OH} up to $\sim 8 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$ [93]. Some key water chemistry controls influencing kinetic parameters under different operating conditions are shown in Table 7.

Kinetic models are frequently extended to fit many species at once (parent+key byproducts) and to incorporate parallel/series reactions for intermediates [72].

Risk-based treatment condition design is informed by the quantification of disinfection by-products (e.g., TCM, TBM) and chlorinated/oxidized intermediates, which are occasionally combined with toxicity evaluation [94]. In the distribution and treatment of drinking water, kinetic characteristics are crucial for forecasting the removal of herbicides and controlling possibly more hazardous or permanent byproducts.

Table 7 Effect of water chemistry and operating conditions on kinetic parameters

Factor	Influence on parameterization	Citations
Temperature	Higher temperature means higher intrinsic k ; Arrhenius behaviour	[72].
Speciation (HClO/ClO ⁻), pH	Strong pH-dependent intrinsic k .	[70, 72].
Background ions, NOM	Can enhance (Cl ⁻ , Br ⁻ via reactive species) or scavenge radicals, lowering k_{obs}	[94].
Oxidant dose (Cl ₂ , O ₃ , PS, H ₂ O ₂)	k_{obs} increases with oxidant (within practical range)	[92, 94].

Critical gaps and Research Challenges

To close the knowledge gaps about the precise fate of herbicides following incursion events into WDS, more research is required. These include pipe material interactions, the impact of residual disinfectants, biofilm-mediated transformations, and hydraulic mixing/dilution effects [67]. Most of the data that is currently accessible comes from laboratory simulations or open aquatic habitats rather than actual distribution networks. Products of transformation provide additional difficulties: regulatory frameworks frequently lag scientific understanding of the dangers associated with these metabolites; some are persistent or more toxic than the main active ingredients [77, 78]. Given the short residence times typical of WDS in comparison to natural waters or treatment facilities, remediation technologies exhibit promise but must be adapted to system-specific constraints [71].

EPANET-MSX for Multi-Species Reactive Transport data Limitations

A flexible engine for modelling several chemical species that interact in WDNs is EPANET-MSX. Although it is a potent tool for multi-species reactive transport (MSRT), its application is severely constrained by performance, input complexity, and data availability. However, using general rate/equilibrium expressions allows users to design arbitrary reaction networks at pipe walls and in bulk water [77, 95]. Additionally, it allows for the simultaneous simulation of numerous species in a single run, such as DBPs, chlorine, water age, haloacetic acids, arsenic species, and taste/odour compounds [89, 96]. For better comparison with experiments, the revised versions add dispersion and enhanced mass-balance reporting [90]. Table 8 shows Key data requirements in some reported EPANET-MSX (MSRT) applications in real-life cases.

Table 8 Key data requirements in some reported EPANET-MSX (MSRT) applications cases

Data type	EPANET-MSX (MSRT) Applications examples	Cita-tions
Boundary/operating data	Temperature, dosing strategy, source concentrations, and mixing of multiple source waters.	[1, 82, 89, 96].
Wall/surface properties	Pipe material, adsorption capacity (e.g., arsenic on iron, wall chlorine demand), and biofilm activity.	[89, 95].
Hydraulic & network	Demands, pressures, pipe diameters/roughness, tanks, controls.	[1, 82].
Kinetic & stoichiometry	Chlorine bulk and wall decay, DBP formation, arsenite oxidation, nicotine reactions, microbial Taste & Odour.	[1, 62, 82, 89].

Model Calibration, Validation, Sensitivity and mass Balance

Any WDS model's quality assurance framework is made up of these four elements. A model is just an untested mathematical exercise without them. Calibration guarantees that the model accurately depicts reality, validation verifies the model's predictive ability, Sensitivity indicates what to measure, identifiability indicates what you can accurately estimate, and mass balance verifies the model's numerical and physical consistency. When combined, they offer the assurance required to make decisions using models [97]. Table 9

Table 9 Steps in ensuring reliability of water quality and hydraulic modelling

Requirement	Description	Minimum standard
Calibration	Modify unknown factors (roughness, reaction rates, sorption coefficients) to match observed data (pressure, flow, concentration).	Make use of a minimum of two separate datasets, such as calibration and verification.
Validation	Test the calibrated model against a separate dataset that was not utilized for calibration.	Verify using literature or bench-scale experiments if field data is not accessible.
Sensitivity Analysis	Identify the parameters that have the most impact on model outputs (e.g., using one-at-a-time, Morris, or Sobol methods).	Determine which parameters need to be measured with high precision.
Identifiability	Determine whether parameters can be uniquely identified from the data at hand (e.g., using correlation analysis).	Simplify the model or gather more data if the parameters are correlated.
Mass Balance	Ensure mass conservation: Input – Output – Reaction – Accumulation = 0 across the network.	Report mass balance error; should be < 1% for numerical models.

summarises how these four elements can be used to improve the quality of water quality and hydraulic modelling.

Conclusion

This paper included a cross-method evaluation, a critical assessment of methodological strengths and limitations, and a future research agenda in contrast to previous, more focused assessments. The review found convergent evidence that the shape and location of leaks, the size and duration of negative pressure episodes, and the interaction with nearby soils and operational transients are the primary factors driving intrusion risk. A systematic technique is needed to develop efficient contamination intrusion, dynamics, and risk assessment methods for herbicide intrusion in water distribution systems. Significant differences still exist, though: estimates of infiltration volumes differ depending on the type of pipe material; health risk propagation through chemical or microbial contaminants is still not consistently quantified; and digital monitoring or early-warning techniques are still not widely used. This entails integrating real-time monitoring systems, comprehending parameter sensitivities, and considering the most relevant specific pollutants. Water utilities can improve public health protection and effectively handle contamination situations by tackling these issues.

A thorough and critical investigation of processes involved in the decay of herbicides and PTPs is important in the prediction of their behaviour and fate in the WDS whenever an intrusion event occurs. There is a wide and persistent gap in the precise fate of herbicides and pesticides after each intrusion event into WDS which includes pipe material interactions, the impact of residual disinfectants, biofilm-mediated transformations, and hydraulic mixing/dilution effects, with parent chemical transformation into more toxic and persistent metabolites. Water conservation and herbicide infiltration are two sides of the same coin. Leaky, inadequately pressurized, and hydraulically sluggish water distribution systems are prone to contamination (allowing herbicide intrusion) and water inefficiency (losing water). On the other hand, a hydraulically engineered, pressurized, and well-maintained network saves water and shields users from herbicide exposure. Moving beyond compartmentalized approaches to water quantity and water quality management, this dual advantage ought to be the cornerstone of utility investment, regulatory policy, and research agendas.

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